

Carisurg MedTech Pathway - Research Assignment

Preliminary Proposal

Project Title: Evaluating Explainable AI-Assisted Emergency Department Triage: A
12-Week Pilot Study

Name: Masele Emmanuel

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Problem Statement

Emergency department (ED) triage is a time-critical process that directly affects patient prioritization, resource allocation, and clinical outcomes. Recent advances in artificial intelligence (AI), machine learning (ML), and natural language processing (NLP) have demonstrated promising improvements in predicting patient acuity, sepsis risk, and hospital disposition (Brann et al., 2024; Chang et al., 2024). However, the literature consistently focuses on retrospective predictive performance rather than how clinicians interact with AI recommendations during routine triage. Furthermore, many studies acknowledge limited external validation and uncertainty regarding performance across different institutions and documentation practices, leaving an important implementation gap between research and clinical deployment.

Summary of Previous Work

Recent research demonstrates significant progress in AI-assisted emergency triage. Chang et al. (2024) combined machine learning and natural language processing to predict emergency department disposition and reported improved predictive performance by incorporating free-text triage information alongside structured clinical data. Similarly, Brann et al. (2024) developed an NLP-based sepsis prediction model capable of identifying sepsis at triage with high discriminative performance, highlighting the value of nursing triage notes in early detection.

Ivanov et al. (2020) showed that machine learning combined with clinical NLP could improve Emergency Severity Index (ESI) acuity assignment compared with routine nurse triage, suggesting potential improvements in consistency and resource allocation. Ivanov et al. (2022) further demonstrated that machine learning could substantially outperform conventional screening methods for detecting sepsis during emergency department triage, while emphasizing the need for prospective evaluation of clinical impact. More recently, Lansiaux et al. (2025) compared NLP, large language model (LLM), and JEPa-based approaches for emergency department triage prediction and reported superior performance from the LLM-based architecture, while noting that ethical transparency and real-world implementation challenges remain unresolved.

Taken together, these studies demonstrate strong algorithmic performance but leave unresolved questions regarding clinician interaction with AI recommendations, explainability, and implementation across diverse clinical environments. These represent practical and meaningful gaps for a short pilot study.

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Proposed Solution

This project proposes a 12-week pilot evaluation of an explainable AI-assisted triage decision support system that supplements, rather than replaces, clinician judgment. The system will generate triage recommendations together with interpretable explanations highlighting the factors contributing to each recommendation. During the pilot, AI recommendations will be compared with clinician-assigned triage decisions to evaluate agreement, consistency, and usability while collecting feedback on clinician trust and workflow integration. The objective is to investigate whether explainable AI can improve decision support without disrupting established emergency department practices.

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